HOTEL RESERVATION MANAGEMENT SYSTEM

1. Introduction and Project Overview

1.1 Background

Accor hotel is a hotel which currently uses manual records to manage its room bookings, guest check-ins, check-outs and billing. This could result in double bookings, delays, inaccurate records and problems with tracking room availability. The proposed Hotel Reservation Management System will automate all of these functions, improve the quality of the records and provide accurate information as to room availability.

1.2 Problem Statement

The present manual booking system is extremely time consuming and error prone. Staff have a real struggle with tracking bookings, updating room status, preparing bills and maintaining guest records. This can all have a negative impact on both customer satisfaction and hotel efficiency. Accor Hotel therefore needs a computerized system to manage its bookings and to improve the quality of its service.

1.3 Project Objectives

The primary aim is to develop a system which will improve hotel reservations and hotel management.

The specific objectives are to:

Cut reservation processing time by at least 50% within three months of going live.

Minimize double bookings via real time room availability updates.

Keep guest, reservation, and payment records up to date.

Automate room allocation, check-in, check-out and billing.

Generate the occupancy and revenue reports for management.

1.4 System type and development approach

System type: It will be predominantly a Transaction Processing System as it will be recording the reservations, room allocations, check-ins, check-outs and payments.

Development approach: Agile. Development will happen in small stages with the hotel staff able to review the features and give feedback at each stage before we complete the system.

2. Project Scope and Requirements

2.1 Project Scope

The project will encompass requirements analysis, system design, development, testing, deployment, user training and documentation. The system will be used by hotel managers, receptionists, accountants and management.

It will handle guest registration, room availability, reservations, check-in and check-out, billing, payment records and reports. The initial version will not include online payment gateway integration or dedicated mobile apps.

2.2 Functional Requirements

The system shall:

Allow authorized staff to log in securely.

Register the guests and keep their details.

Record the reservation, the date, the room type and the guest details.

Show available and occupied rooms and prevent conflicting bookings.

Manage room allocation, check-in and check-out.

Figure out bills and write down payments.

Generate receipts and reservation/occupancy/revenue reports.

Allow admins to manage room rates and staff accounts.

2.3 Non-Functional Requirements

The system must be user-friendly, robust and accessible via modern web browsers. It must respond to standard requests within 3 seconds, provide for access controls to protect guest data and have regular database backups.

2.4 Stakeholders

The stakeholders are the hotel owner, the hotel manager, the receptionists, the accountants, the guests, the systems analyst, the software developers and the system administrator.

3. FEASIBILITY STUDY

3.1 Economic Feasibility

Estimated project cost is KSh 1.6m. The expected benefits are reduced admin, fewer booking errors, better room occupancy and faster service.

3.2 Technical Feasibility

The system can be built using HTML, CSS, JavaScript, an appropriate server side language and MySQL.

3.3 Operational Feasibility

The system is designed to simplify day to day hotel operations and reduce the reliance on paper. Staff will be trained prior to implementation and feedback from users will be gathered during the testing phase.

3.4 Schedule Feasibility

The project is expected to take 12 weeks. There will be weekly progress reviews to pick up any issues and deal with any delays.

3.5 Legal, Contractual, and Organizational Feasibility

The system will be designed to be compliant with applicable Kenyan data protection legislation and guest data will be restricted to those authorized to have access. Software licenses and hosting contracts will be reviewed. Management support and cooperation from the various hotel departments will be essential to the success of the project.

4. Deliverables and Timeline

4.1 Project Deliverables

The project will produce:

Project Initiation Document and Feasibility Study.

Software requirements specification.

System and database design documents.

A functional Hotel Reservation Management System.

A set up database and deployment environment.

System testing report.

User manual and training materials.

Final project report and deployed system.

4.2 Project Timeline

The project runs from 9 November 2026 to 29 January 2027.

Activity	Duration	Completion date
Initiation and feasibility study	1 week	13 Nov 2026
Requirements analysis	2 weeks	27 Nov 2026
System design	2 weeks	11 Dec 2026
System development	3 weeks	1 Jan 2027
Testing and corrections	2 weeks	15 Jan 2027
Deployment and training	1 week	22 Jan 2027
Evaluation and handover	1 week	29 Jan 2027

5. PROJECT BUDGET

The projected project budget is KSh 1.6 million.

Budget item	Cost (KSh)
Requirements analysis	150,000
System and database design	180,000
Software development	450,000
Computers and networking	250,000
Hosting and deployment	80,000
Testing and quality assurance	120,000
Training and documentation	90,000

Initial maintenance and support 120,000

Contingency reserve 160,000

Total 1,600,000

6. PROJECT RISKS AND MANAGEMENT

The main risks and mitigation measures are:

Risk Mitigation measure

Double bookings: validate room availability prior to confirming the booking.

Unauthorized access: use secure login and role based permissions.

Data loss: maintain regular backups and test your recovery.

Staff resistance: provide training and ongoing support.

Development delays: monitor progress on a weekly basis and prioritize the key features.

Budget overruns: track expenditure and formally approve any changes.

The project manager will keep a risk register and will review the risks at the weekly meetings.

7. Project reporting and communication

The project manager will produce weekly progress reports for the hotel manager. These will cover completed work, work in progress, milestone progress, expenditure, risks and any issues which require a decision. Meetings will be held at the end of each major stage of development. Any significant delays and any security incidents will be reported as and when they occur.

8. PROJECT ACCEPTANCE AND CONCLUSION

The system will be accepted upon implementation of the agreed requirements, all critical defects being rectified, user acceptance testing having been approved, staff training having been completed and the system having been successfully deployed. Final handover will be approved by the hotel manager.